

Question 1 (18 Marks)**Marks**

(a) Factorise fully

(i) $16x^3 - 54y^3$ **2**

(ii) $x^2 - y^2 + y - x$ **2**

(iii) $a^5 - 2a^3 + a$ **3**

(b) Simplify $\frac{1}{y-x} + \frac{x}{(x-y)^2}$ **3**

(c) Express $2.0\dot{3}9$ as a rational number in its simplest form. **3**

(d) Express $2\sqrt{27} - \frac{\sqrt{12}}{2} + 3\sqrt{48}$ in its simplest form. **2**

(e) Solve the equations $x - y = 6$ and $x^2 + y^2 = 18$ simultaneously **3**
and explain the geometric significance of your answer.

Question 2 (18 Marks) (START QUESTION IN A NEW BOOKLET)

(a) Solve

(i) $|x - 3| = 11$ **2**

(ii) $|2x - 11| = 3x - 4$ **3**

(iii) $|1 - 2x| \leq 5$ **2**

(iv) $\frac{1-y}{6} - \frac{y-2}{3} = \frac{y+1}{4}$ **2**

(b) Express $1 + \frac{2}{\sqrt{3}-1}$ in the form $a + b\sqrt{3}$ where a and b are rational. **3**

Question 2 Continued.**Marks**

(c) If $x - y + \sqrt{x + y} = \sqrt{6}$, find the value of x and y . **2**

(d) Solve $\frac{x}{x-1} < -2$ **4**

Question 3 (18 Marks) (START QUESTION IN A NEW BOOKLET)

(a) If $f(x) = x^2 + 2$ and $g(x) = 2x + 3$, find the value of $f(g(3))$. **2**

(b) Write down the natural domain of $f(x) = \frac{1}{\sqrt{x^2 - 9}}$ **1**

(c) Sketch the function $y = \frac{2x-3}{x-1}$, stating the equations of its vertical and horizontal asymptotes? **5**

(d) Noting any restrictions on the domain of the function, make a neat sketch of the graph of **2**

$$y = \frac{x^2 - 9}{x + 3}$$

(e) Solve $|x-2| + |x+1| = 3$. **3**

(f) Sketch the graphs of $y = |x+2|$ and $y = |x-4|$ on the same set of axes and use your graphs to solve $|x+2| \leq |x-4|$. **5**

Question 4 (18 Marks) (START QUESTION IN A NEW BOOKLET)

(a) Without using a calculator, prove that $\frac{1 - \tan^2 30^\circ}{1 + \tan^2 30^\circ} = \cos 60^\circ$. **3**

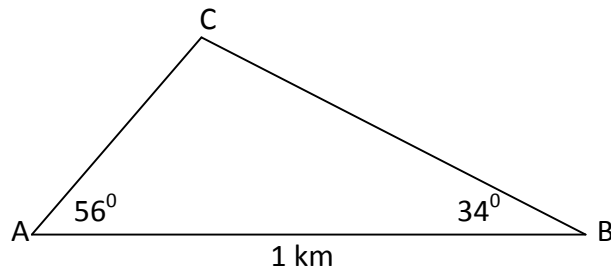
(b) If θ is acute and $\sec \theta = \frac{\sqrt{5}}{2}$, find the exact value of: **3**

(i) $\sin \theta$

(ii) $\tan \theta$

Question 4 Continued.**Marks**

- (c) From two observation points A and B 1 km apart, a balloon C is observed at angles of elevation of 56° and 34° respectively. Find to the nearest metre, the height of the balloon above the ground. **4**



- (d) Find the exact value of $\frac{\sin 135^\circ - \cos 240^\circ}{\sin 225^\circ + \cos 120^\circ}$. **2**
- (e) If $\tan \theta = -\frac{5}{12}$ and $270^\circ < \theta < 360^\circ$, find the exact value of $\cos \theta$. **2**
- (f) Prove $\frac{1}{\sec \theta + \tan \theta} = \frac{\cos \theta}{1 + \sin \theta}$. **2**
- (g) Prove $\frac{2 \cos^2 \theta - 1}{\cos^2 \theta - \sin^2 \theta} = 1$. **2**

END OF PAPER

Q1 (a) (i) $2(8x^3 - 27y^3) = 2(2x-3y)(4x^2+6xy+9y^2)$

(ii) $(x-y)(x+y) - (x-y) = (x-y)(x+y-1)$

(iii) $a(a^4 - 2a^2 + 1) = a(a^2 - 1)^2 = a(a-1)^2(a+1)^2$

(b) $-\frac{1}{x-y} + \frac{x}{(x-y)^2} = \frac{-(x-y) + x}{(x-y)^2} = \frac{y}{(x-y)^2}$

(c) Let $x = 2.039$

$\therefore 10x = 20.399$

$\therefore 9x = 18.36$

$x = \frac{1836}{900}$

$= 2\frac{1}{25}$

(d) $6\sqrt{3} - \sqrt{3} + 12\sqrt{3} = 17\sqrt{3}$

(e) $y = x - 6$
 $\therefore x^2 + (x-6)^2 = 18$

$x^2 + x^2 - 12x + 36 = 18$

$2x^2 - 12x + 18 = 0$

$x^2 - 6x + 9 = 0$

$(x-3)^2 = 0$

$\therefore x = 3$

$y = -3$

Since there is only one solution the line is a tangent to the circle.

Q2

(a) (i) $x - 3 = 11$

$x = 14$

$x - 3 = -11$

$x = -8$

(ii) $2x - 11 = 3x - 4$

$x = -7$

$2x - 11 = -(3x - 4)$

$2x - 11 = -3x + 4$

$5x = 15$

$x = 3$

Testing solutions

$x = -7$: LHS = 25, RHS = -25

$x = 3$: LHS = 5, RHS = 5

 $\therefore x = -7$ is not a solution $\therefore$ Solution is $x = 3$

(iii) $1 - 2x \leq 5$

$-2x \leq 4$

$x \geq -2$

$1 - 2x \geq -5$

$-2x \geq -6$

$x \leq 3$

 $\therefore$ Solution is $-2 \leq x \leq 3$

(iv) $2(1 - y) - 4(y - 2) = 3(y + 1)$

$2 - 2y - 4y + 8 = 3y + 3$

$10 - 6y = 3y + 3$

$9y = 7$

$y = \frac{7}{9}$

(b)

$\frac{\sqrt{3} - 1 + 2}{\sqrt{3} - 1}$

$= \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \cdot \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$

$= \frac{(\sqrt{3} + 1)^2}{3 - 1}$

$= \frac{3 + 2\sqrt{3} + 1}{2}$

$= \frac{4 + 2\sqrt{3}}{2}$

$= 2 + \sqrt{3}$

$$\begin{aligned}
 (c) \quad & \left. \begin{aligned} x - y &= 0 \\ x + y &= 6 \end{aligned} \right\} \checkmark \\
 & 2x = 6 \\
 & \left. \begin{aligned} x &= 3 \\ y &= 3 \end{aligned} \right\} \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (d) \quad & \frac{(x-1)^2}{(x-1)} < -2 \cdot (x-1)^2 \quad \checkmark \\
 & x(x-1) < -2(x-1)^2 \\
 & 2(x-1)^2 + x(x-1) < 0 \quad \checkmark \\
 & (x-1)(2x-2+x) < 0 \\
 & (x-1)(3x-2) < 0 \quad \checkmark \\
 & \therefore \frac{2}{3} < x < 1 \quad \checkmark
 \end{aligned}$$

Q3

(a) $g(3) = 9$ ✓

$\therefore f(9) = 81 + 2 = 83$ ✓

(b) $x < -3, x > 3$ ✓

(c) Vertical asymptote is $x = 1$ ✓

Horizontal asymptote:

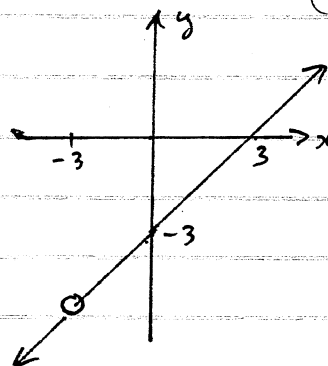
$$y = \frac{2 - \frac{3}{x}}{1 - \frac{1}{x}}$$
 ✓

as $x \rightarrow \infty, y \rightarrow \frac{2}{1} = 2$

$\therefore y = 2$ ✓

(d) Function is undefined for $x = -3$

Otherwise $y = \frac{(x-3)(x+3)}{(x+3)} = x-3$ ✓



(e) (i) $(x-2)^2 + (y-3)^2 = 16$ ✓

(ii)

Solve

$y = x$

$(x-2)^2 + (y-3)^2 = 16$

as $y = x$ passes through the points of intersection ✓

$(x-2)^2 + (x-3)^2 = 16$

$x^2 - 4x + 4 + x^2 - 6x + 9 = 16$

$2x^2 - 10x + 3 = 0$ ✓

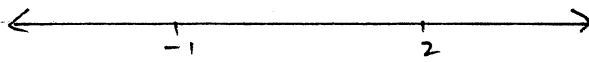
$$x = \frac{10 \pm \sqrt{100 - 24}}{4}$$

$$= \frac{10 \pm \sqrt{124}}{4}$$

$$\therefore x = \frac{5 \pm \sqrt{31}}{2}$$

$$y = \frac{5 \pm \sqrt{31}}{2}$$
 ✓

e (*)

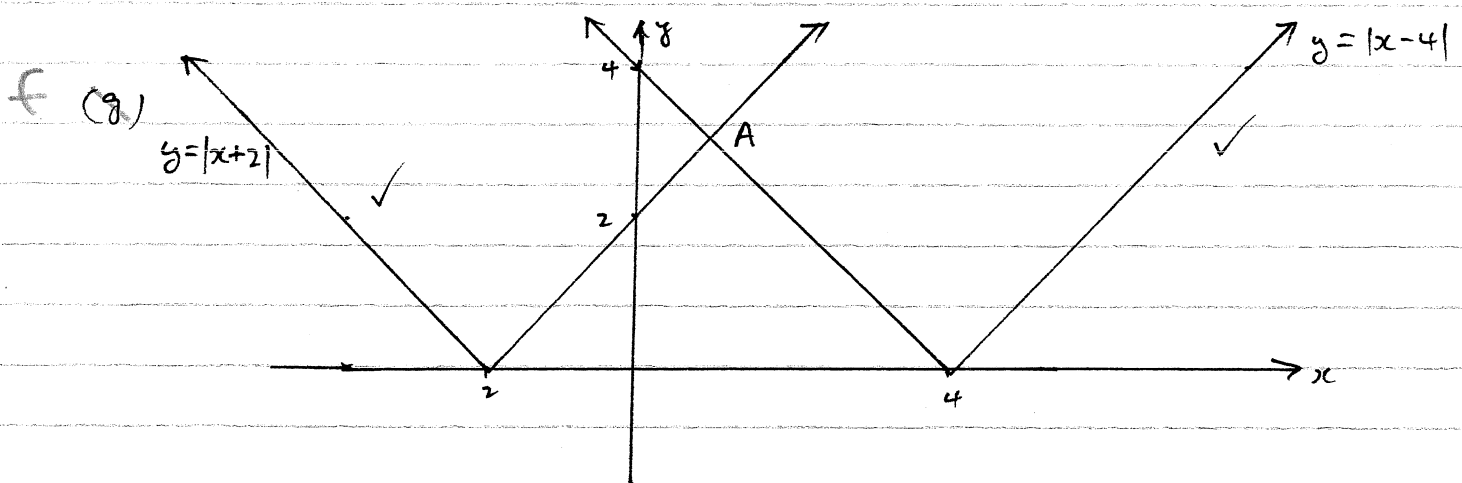


Defining values of x are -1 and 2 ✓

For $-1 < x < 2$

$$\begin{aligned}\text{Expression} &= -(x-2) + (x+1) \\ &= -x + 2 + x + 1 \\ &= 3\end{aligned}$$

∴ $-1 < x < 2$ ✓



To find the point A

$$\begin{aligned}\text{Solve } \left. \begin{aligned} y &= -(x-4) \\ y &= x+2 \end{aligned} \right\} & \checkmark \\ y &= -x+4 \\ y &= x+2 \\ -x+4 &= x+2 \\ 2x &= 2 \\ x &= 1\end{aligned}$$

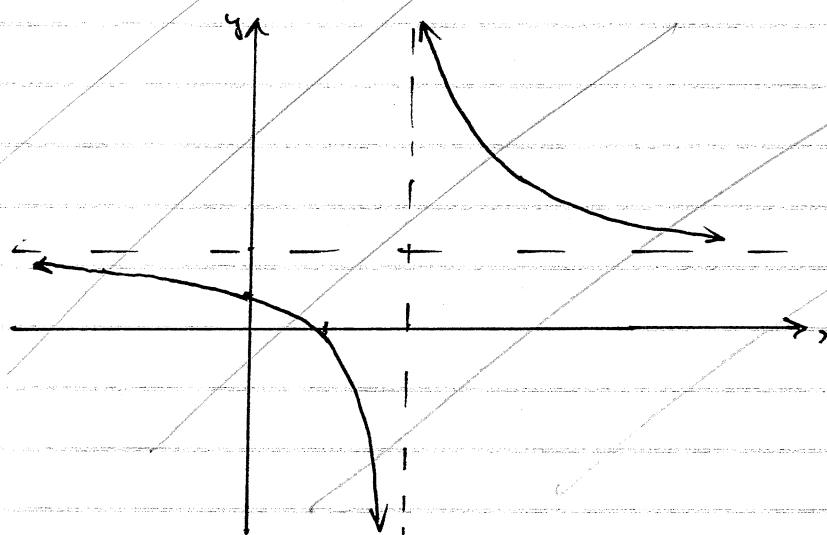
∴ From the graph solution is $x \leq 1$ ✓

Q4

(a)

$$(y-1) = \frac{1}{(x-2)}$$

✓



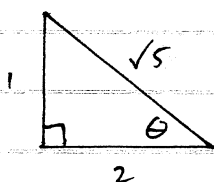
(a) (b)

$$\text{LHS} = \frac{1 - \left(\frac{1}{\sqrt{3}}\right)^2}{1 + \left(\frac{1}{\sqrt{3}}\right)^2} = \frac{1 - \frac{1}{3}}{1 + \frac{1}{3}} = \frac{\frac{2}{3}}{\frac{4}{3}} = \frac{1}{2}$$

$$\text{RHS} = \cos 60^\circ = \frac{1}{2}$$

∴ LHS = RHS Hence result. ✓

(b) (c)



Using Pythagoras, missing side is 1 ✓

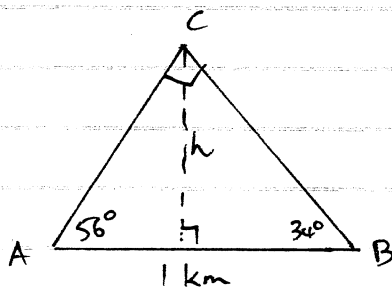
$$\therefore \text{(i)} \sin \theta = \frac{1}{\sqrt{5}} \quad \checkmark$$

$$\text{(ii)} \tan \theta = \frac{1}{2} \quad \checkmark$$

$$\therefore \sqrt{(1 - \sin^2 \theta)(1 + \tan^2 \theta)} = \sqrt{\left(1 - \frac{1}{5}\right)\left(1 + \frac{1}{4}\right)} \quad \checkmark$$

$$= \sqrt{\frac{4}{5} \times \frac{5}{4}} = \sqrt{1} = 1$$

C (d)



$$\angle ACB = 90^\circ \quad \checkmark$$

$$\therefore \cos 56^\circ = \frac{AC}{1000} \quad \checkmark$$

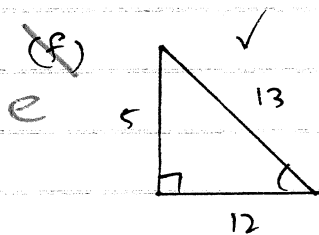
$$AC = 1000 \cos 56^\circ$$

$$\sin 56^\circ = \frac{h}{AC} \quad \checkmark$$

$$\therefore h = (1000 \cos 56^\circ) \sin 56^\circ$$

$$\therefore h = 464 \text{ m to nearest metre} \quad \checkmark$$

$$\begin{aligned}
 d) \quad & \frac{\frac{1}{\sqrt{2}} - (-\frac{1}{\sqrt{2}})}{-\frac{1}{\sqrt{2}} + (-\frac{1}{\sqrt{2}})} = \frac{\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}}{-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}} \\
 & = \frac{\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}}{-(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}})} = -1 \quad \checkmark
 \end{aligned}$$



Using the acute equivalent of θ
and the fact that $\cos \theta$ is
positive in the range

$$\cos \theta = \frac{12}{13} \quad \checkmark$$

$$\begin{aligned}
 f) \quad \text{LHS} &= \frac{1}{\sec \theta + \tan \theta} = \frac{1}{\frac{1}{\cos \theta} + \frac{\sin \theta}{\cos \theta}} \\
 &= \frac{1}{\frac{1 + \sin \theta}{\cos \theta}} \times \cos \theta \\
 &= \frac{\cos \theta}{1 + \sin \theta} \\
 &= \text{RHS}
 \end{aligned}$$

$$\begin{aligned}
 g) \quad \text{LHS} &= \frac{2 \cos^2 \theta - 1}{\cos^2 \theta - \sin^2 \theta} \\
 &= \frac{2 \cos^2 \theta - 1}{\cos^2 \theta - (1 - \cos^2 \theta)} \\
 &= \frac{2 \cos^2 \theta - 1}{\cos^2 \theta + \cos^2 \theta - 1} \\
 &= \frac{2 \cos^2 \theta - 1}{2 \cos^2 \theta - 1} \\
 &= 1 \\
 &= \text{RHS}
 \end{aligned}$$